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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Deep Learning - IIT Ropar (course)

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Week 5 : Assignment 5

Assignment not submitted

Due date: 2026-02-25, 23:59 IST.

Information for questions 1 to 5

Case Study: Content Recommendation Dynamics in a Streaming Platform

A video streaming platform categorizes its content into three major genres: Drama (D), Comedy (C), and Documentary (O). Based on historical viewing data, the platform observes that users frequently switch between these genres from one day to the next. To analyze long-term viewing behavior, the platform models these transitions using a probabilistic framework.

The transition behavior is represented using the matrix:

$$M = \begin{bmatrix} 0.6 & 0.2 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.2 & 0.3 & 0.4 \end{bmatrix}$$

An initial distribution of users across the three categories is represented by a vector v_0 . The platform repeatedly applies this transition model to understand how content popularity stabilizes over time.

1) Which property of matrix M makes it suitable for modeling user transitions?

1 point

- ☐ It is diagonal and simplifies computation
- ☐ Each row sums to zero, ensuring conservation
- ☐ It preserves probability distributions across categories

Course outline

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How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

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Week 5 ()

- ☐ Eigenvalues and Eigenvectors (unit?unit=71&lesson=72)
- ☐ Linear Algebra : Basic Definitions (unit?unit=71&lesson=73)
- ☐ Eigenvalue Decompositon (unit?unit=71&lesson=74)
- ☐ Principal Component Analysis and its Interpretations (unit?unit=71&lesson=75)
- ☐ PCA : Interpretation 2 (unit?unit=71&lesson=76)
- ☐ PCA : Interpretation 3 (unit?unit=71&lesson=77)
- ☐ PCA : Interpretation 3 (Contd.) (unit?unit=71&lesson=78)
- ☐ PCA : Practical Example (unit?unit=71&lesson=79)
- ☐ Singular Value Decomposition (unit?unit=71&lesson=80)
- ☐ Lecture Material for Week 5 (unit?unit=71&lesson=81)
- ☐ Week 5 Feedback Form:Deep Learning - IIT Ropar!! (unit?unit=71&lesson=234)
- ☐ Quiz: Week 5 : Assignment 5

☐ It contains distinct eigenvalues for each genre

2) What can be inferred about the dominant eigenvalue of matrix M ?

1 point

- ☐ It is greater than 1 due to amplification effects
- ☐ It is equal to 1 as total probability is conserved
- ☐ It is negative because probabilities decay
- ☐ It varies depending on the initial user distribution

3) If the platform repeatedly applies the transformation $v_{k+1} = Mv_k$, what will eventually happen?

1 point

- ☐ User distribution oscillates indefinitely
- ☐ Users permanently concentrate in one genre
- ☐ The distribution converges to a steady state
- ☐ The magnitude of the vector increases over time

4) If the initial distribution of users is changed, will the final long-term distribution also change?

1 point

- ☐ Yes, early preferences permanently bias the outcome
- ☐ No, long-term behavior depends only on the transition model
- ☐ Only if the matrix is symmetric
- ☐ Only if complex eigenvalues are present

5) In this scenario, what does the dominant eigenvector of matrix M represent?

1 point

- ☐ A temporary surge in content popularity
- ☐ The genre with the fastest short-term growth
- ☐ The long-term proportion of users in each genre
- ☐ A randomly selected direction in vector space

Information for questions 6 to 10

Case Study: Stability of Sensor Signals in an IoT Monitoring System

An IoT-based environmental monitoring system tracks two independent sensor signals, such as temperature and humidity. Due to hardware constraints and environmental factors, each update slightly reduces the signal strength. This behavior is modeled using a linear transformation applied at every time step.

The update matrix is given by:

Week 6 ()

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$$A = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.4 \end{bmatrix}$$

The initial sensor readings are represented by a vector x_0 . Engineers analyze how these readings evolve over time to assess system stability.

6) What are the eigenvalues of matrix A ?

1 point

- ☐ 0.7 and 0.4
- ☐ 1 and 0
- ☐ -0.7 and -0.4
- ☐ 0.28 and 0.16

7) What happens to the sequence $x_k = A^k x_0$ as $k \rightarrow \infty$?

1 point

- ☐ It grows without bound
- ☐ It oscillates between fixed values
- ☐ It converges to the zero vector
- ☐ It stabilizes at a non-zero constant

8) Which sensor signal diminishes more rapidly over time?

1 point

- ☐ The first signal due to higher attenuation
- ☐ The second signal due to faster decay
- ☐ Both signals decay at the same rate
- ☐ The decay rate cannot be determined

9) What does this behavior indicate about the long-term system response?

1 point

- ☐ Signals continuously amplify
- ☐ Noise dominates and destabilizes the system
- ☐ Signals gradually stabilize and fade
- ☐ The system becomes chaotic over time

10) If one eigenvalue of matrix A were greater than 1, what would be the consequence?

1 point

- ☐ The system output becomes constant
- ☐ Signal values grow exponentially in that direction
- ☐ All signal components eventually vanish
- ☐ Eigenvectors no longer exist

Information for questions 11 to 15

Case Study: Feature Compression Using Principal Component Analysis

A university collects extensive student interaction data from an online learning platform, including time spent on lectures, number of clicks, assignment attempts, and resource usage. Many of these features are highly correlated, making analysis and visualization challenging. To reduce dimensionality while preserving important patterns, the analytics team applies Principal Component Analysis (PCA).

11) What is the fundamental mathematical operation underlying PCA?

1 point

- ☐ Iterative gradient optimization
- ☐ Eigen-decomposition of the covariance matrix
- ☐ Random sampling of feature subsets
- ☐ Simple feature normalization

12) Why are principal components orthogonal to each other?

1 point

- ☐ To reduce computational complexity
- ☐ To increase feature correlation
- ☐ Due to properties of symmetric covariance matrices
- ☐ To preserve original data scale

13) Which principal components are typically removed during dimensionality reduction?

1 point

- ☐ Components with high variance
- ☐ Components with large eigenvalues
- ☐ Components with low variance
- ☐ Components closest to the mean

14) What does a principal component represent?

1 point

- ☐ A single observed feature
- ☐ A nonlinear transformation of inputs
- ☐ A linear combination of original features
- ☐ An average behavior pattern

15) While applying dimensionality reduction on the student activity dataset, the analytics team decides to use Singular Value Decomposition (SVD) directly on the data matrix instead of computing the covariance matrix first. What is the primary advantage of this approach?

1 point

- ☐ SVD ensures that all transformed features become statistically independent
- ☐ SVD can be applied effectively even when the no. of features exceeds the samples
- ☐ SVD automatically removes noise without selecting components
- ☐ SVD converts nonlinear relationships into linear representations

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers